

Translated

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Chinese Remainder Theorem

The Chinese Remainder Theorem (which will be referred to as CRT in the rest of this article) was discovered by Chinese mathematician Sun Zi.

Formulation

Let $m = m_1 \cdot m_2 \cdot \dots \cdot m_k$, where m_i are pairwise coprime. In addition to m_i , we are also given a system of congruences

$$\left\{ \begin{array}{l} a \equiv a_1 \pmod{m_1} \\ a \equiv a_2 \pmod{m_2} \\ \vdots \\ a \equiv a_k \pmod{m_k} \end{array} \right.$$

where a_i are some given constants. The original form of CRT then states that the given system of congruences always has *one and exactly one* solution modulo m .

E.g. the system of congruences

$$\left\{ \begin{array}{l} a \equiv 2 \pmod{3} \\ a \equiv 3 \pmod{5} \\ a \equiv 2 \pmod{7} \end{array} \right.$$

has the solution 23 modulo 105 , because $23 \bmod 3 = 2$, $23 \bmod 5 = 3$, and $23 \bmod 7 = 2$. We can write down every solution as $23 + 105 \cdot k$ for $k \in \mathbb{Z}$.

Corollary

A consequence of the CRT is that the equation

$$x \equiv a \pmod{m}$$

is equivalent to the system of equations

$$\left\{ \begin{array}{l} x \equiv a_1 \pmod{m_1} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{array} \right.$$

(As above, assume that $m = m_1 \cdot m_2 \cdot \dots \cdot m_k$ and m_i are pairwise coprime).